AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Final Examination
Fall Semester 2025/2026

Date: 21 /Jan. /2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

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Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Part 1: Multiple Choice Questions (MCQ) [20 marks]

1- Which method is used to extract a substring from a string starting at a given index?

a. Slice()
b. Substring()
c. Extract()
c. Cut()

2- What is the output of "C#Exam".ToUpper()?

a. C#exam
b. C#EXAM
c. c#EXAM
d. C#Exam

3- Which method of List<T> removes all elements?

a. Clear()
b. Delete()
c. Reset()
d. Empty()

4- Which of the following keywords is used to define a method that must be overridden in deri classes?

a. static
b. virtual
c. sealed
d. abstract

5- What is the purpose of a constructor in a class?

a. Destroy the object
b. Initialize new objects
c. Access static fields
d. Prevent method override

6- What is the purpose of this keyword in C#?

a. To define a class
b. To refer to the current instance
c. To delete an object
d. To call a static method

7- Which class allows you to write text files line by line?

a. FileReader
b. FileWriter
c. StreamWriter
d. TextWriter

8- Which of the following is NOT a valid access modifier in C#?

a- public
b- hidden
c- protected
d- private

9- What is the purpose of the virtual keyword in C#?

a- To prevent inheritance
b- To make a method static
c- To allow method overriding
d- To restrict access

10- What is polymorphism?

a- Code reuse
b- Multiple forms of an object
c- Data hiding
d- Memory management

Part 2: Short Questions [20 marks]

Q1- Briefly explain the difference between List<T> and ArrayList.

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Q2- Assume the following code is error free. What is the output of the following program?

```
public class Device {
    public virtual string Name => "Generic Device";
}
public class Phone : Device {
    public override string Name => "Smartphone";
}
Device d = new Phone();
Console.WriteLine(d.Name);
```

Q3- Assume the following code is error free. What is the output of the following program?

```
string s = "Programming";
Console.WriteLine(s.IndexOf('g'));
```

Q4- Assume the following code is error free. What is the output of the following program?

```
var numbers = new List<int> { 10, 3, 7, 1 };
numbers.Reverse();
Console.WriteLine(numbers[1]);
```

Q5- Check the following code and if it is correct, specify how many times the following loop will run?

```
int i=10,x=0;
      do
      {
         x = x + i;
         i=i+5;
         Console.WriteLine(x);
      } while (i < 12);
```

Part 3: Problem Solving Questions [60 marks]

Q1- Develop the following classes to model product management system. [30 marks]

Base Class (Product):

1. **Properties:**
   - **Id (int, read-only)**
   - **Name (string)**
   - **BasePrice (decimal)**
   - **Stock (int, protected)**
2. **Methods:**
   - **Restock(int quantity): Increases stock**
   - **Sell(int quantity): Decreases stock (throws exception if insufficient stock)**
   - **GetPrice(): Returns BasePrice**
   - **GetProductInfo(): Returns formatted string with all product details**

Derived Class (DiscountedProduct):

1. **Additional Property:**
   - **DiscountRate (decimal, 0-1 range)**
2. **Special Behavior:**
   - **Overrides GetPrice() to return discounted price**

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Answer ONLY two questions of the following questions.

Q2- Write a C# method to count the frequency of each character in a string. [15 marks]

Sample I/O

Input: "hello"

Output: h:1, e:1, l:2, o:1

Q3- Write a C# program that merges two arrays and sorts the result. [15 Marks]

Sample I/O

Input: [2, 4, 6], [1, 3, 5]
Output: [1, 2, 3, 4, 5, 6]

Q4- Write a C# program to calculate the sum of elements for each column in a 2D array.. [15 Marks]

Sample I/O:

```
int[,] arr = {
   {1, 2, 3},
   {4, 5, 6}
 };
```

Output: Column sums: 5, 7, 9

End of questions

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